Problem 15.17

You have \$10,000 down payment on a \$20,000 car. The dealer offers you the following two options:

(a) paying the balance with end-of-month payments over the next three years at $i^{(12)} = 0.12$;

(b) a reduction of \$500 in the price of the car, the same down payment of \$10,000, and bank financing of the balance after down payment, over 3 years with end-of-month payments at $i^{(12)} = 0.18$.

Which option is better?

Solution

We compare the required monthly payments under each option. Since payments are made monthly for 3 years, the number of payments is

$$3(12) = 36.$$

Option (a)

The amount financed is

$$20000 - 10000 = 10000.$$

The nominal annual interest rate convertible monthly is $i^{(12)} = 0.12$, so the monthly interest rate is

$$j = \frac{0.12}{12} = 0.01.$$

If the monthly payment is X, then

$$10000 = X a_{\overline{36}|}.$$

Thus,

$$\begin{aligned} X &= \frac{10000}{a_{\overline{36}|}} \\ &= \frac{10000}{\dfrac{1-(1.01)^{-36}}{0.01}} \\ &\approx 332.1431. \end{aligned}$$

Option (b)

With the \$500 price reduction, the new price of the car is

$$20000 - 500 = 19500.$$

After the same \$10,000 down payment, the amount financed is

$$19500 - 10000 = 9500.$$

The nominal annual interest rate convertible monthly is $i^{(12)} = 0.18$, so the monthly interest rate is

$$j = \frac{0.18}{12} = 0.015.$$

If the monthly payment is X, then

$$9500 = Xa_{\overline{36}|}.$$

Thus,

$$\begin{aligned} X &= \frac{9500}{a_{\overline{36}|}} \\ &= \frac{9500}{\dfrac{1-(1.015)^{-36}}{0.015}} \\ &\approx 343.4478. \end{aligned}$$

Since option (a) has the lower monthly payment, the better option is

$$\boxed{\text{Option (a)}}.$$